# ABSTRACT

Radiation monitoring and mapping are fundamental to nuclear safety, environmental monitoring and emergency response. This Summer Internship Report presents the research and development work carried out on the project **"RADMAP – RADiation MAPing – GPS Enabled Radiation Mapping Prototype on Raspberry Pi-5 (3D Visualization)"** at the Amity Institute of Nuclear Science & Technology (AINST). The first goal of the project was to build a portable and intelligent radiation mapping system that would provide real time radiation measurements and geographical coordinates, environmental parameters and current advanced visualization for effective radiation monitoring and future source localization.

During the 47 day internship, a wide range of research and development projects were conducted on embedded systems, geospatial mapping, radiation detection and data visualization. This work included creating and preprocessing a radiation dataset, uploading the data to Kaggle, and using the data to create interactive 2D radiation maps. A real time GPS acquisition module was designed using **GlobalSat BU-353N** GPS receiver, which was attached to the Raspberry Pi-5 to acquire latitude, longitude, UTC time, temperature and humidity data and were used for continuous environmental monitoring and data processing.

The project also included integration and testing of a **CNSPEC Multi-Channel Analyzer (Gamma Spectrometer)** with standard radioactive sources like **Cobalt-60 (Co-60)** and **Cesium (Cs-137).** Radiation count rates, spectral peaks and detector responses were measured and analyzed with success and dynamic radiation intensity maps were generated. In order to address the limitation of the conventional **2D heatmaps**, research was carried out for radioactive source localization on the **3D Digital Twin** technology. To visualize the monitored environment in real time, a real-time visualization system was developed by synchronizing the live image of the camera with GPS coordinates and radiation data, so that the three-dimensional image of the environment monitored by the camera could be obtained.

Moreover, a full backend framework was created to allow for integration of sensor communication, *GPS telemetry*, radiation processing, and visualization modules into a single system. A color grading algorithm, based on the technology used in night vision imaging, was used to illustrate different radiation intensity levels by using different color scales in the live camera view, and thus provide intuitive radiation hotspot detection. Additionally, a

prototype model in the form of a 3D printable model was designed with the final hardware architecture in mind for future prototype development of **RADMAP**.

Success in implementing and validating the prototype of **RADMAP** confirmed the possibility of implementing radiation mapping in real time based on GPS navigation, embedded computing, and intelligent visualization methods. **Dr. Abhishek Yadav** & **Prof. (Dr.) Alpana Goel**, **Director of AINST**, appreciated and approved the project for further development towards next generation radiation mapping, autonomous source localization and advanced Digital Twin technologies. The internship was a great practical experience with embedded systems, IoT, geospatial analytics, radiation instrumentation and research-oriented software development, and helped to develop innovative solutions in the areas of nuclear safety and environmental monitoring.